

Aishwarya Hareesh Rao

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M.S. Electrical Engineering @ Arizona State University · Embedded Systems & TinyML · Real-Time Signal Processing & VLSI ·

Seeking Fall 2026 internship opportunities

EDUCATION

Arizona State University

M.S. Electrical Engineering

Tempe, Arizona · Jan 2025 – Dec 2026

Visvesvaraya Technological University

B.E. Electronics & Communication Engineering

Belagavi, India · Aug 2020 – Aug 2024

TECHNICAL SKILLS

Languages: Python, C, C++, Embedded C, MATLAB, Verilog

AI / ML & TinyML: TensorFlow, TFLite Micro, PyTorch, Scikit-Learn, MFCC Feature Extraction, Model Quantization, CNNs, Experimental Design

Embedded & Real-Time: Arduino Nano 33 BLE Sense, ARM Cortex MCUs, PDM Microphones, ROS2, Gazebo, Sensor Integration, Real-Time Signal Processing

VLSI & EDA: Cadence Virtuoso, HSPICE, StarRC Extraction, DRC/LVS Verification, Standard-Cell Layout, Post-Layout Timing Analysis

Analytics & Tools: Anomaly Detection, Time-Series Analysis, Closed-Loop Control, Root-Cause Analysis, Git, Linux, Jupyter, CUDA, VS Code

WORK EXPERIENCE

Centre for Artificial Intelligence & Robotics (DRDO-CAIR)

Project Intern

Bengaluru, India

Sep 2023 – Nov 2023

- **Reduced subsystem integration cycle time** by applying structured ROS2 + Gazebo bring-up and debugging workflows — validated robotic subsystems end-to-end with **zero rework cycles** post-qualification.
- **Improved sensor measurement stability and accuracy** by calibrating and fine-tuning multiple onboard sensors via root-cause deviation analysis — eliminated systematic noise artifacts affecting downstream perception and control modules.

PROJECTS

Audio-Based Keyword Spotting System — End-to-End TinyML Pipeline

Aug – Dec 2025 · ASU

- **Achieved real-time on-device keyword inference** by building a complete TinyML pipeline (250+ sample dataset → MFCC extraction → noise augmentation → quantized model) deployed on **Arduino Nano 33 BLE Sense** — validated within resource-constrained hardware memory and latency limits.
- **Maximized classifier generalization under real-world acoustic variability** by curating a balanced dataset with noise augmentation and normalization — demonstrated full AI lifecycle ownership from raw sensor input to embedded decision output with **zero external inference dependency**.

4-to-1 Integrate-and-Fire (IF) Neuron Design — VLSI Standard-Cell Layout + Post-Layout Timing

Jan – May 2025 · ASU

- **Achieved DRC/LVS-clean sign-off with zero violations** by designing standard-cell layouts (adders, flip-flops, neuron blocks) in **Cadence Virtuoso** with disciplined routing and layer constraints — delivered a pipelined, synchronized architecture stable under full parasitic load.
- **Quantified critical-path delay and timing margins** by running post-layout **HSPICE simulations with StarRC extraction** — produced a verified timing report analogous to inline electrical evaluation in production semiconductor fabs, enabling confident design closure.

Driver Distraction & Drowsiness Detection — Multi-Sensor Monitoring with Closed-Loop Alerts

Jan – Aug 2024 · RVITM

- **Detected abnormal driver patterns at low false-positive rates** by extracting **PERCLOS, EAR, and gaze-deviation metrics** from a multi-sensor system and applying threshold-based time-series analysis on head-pose angle deviations — surfaced event signatures reliably in real-time.
- **Enabled automated safety response** by developing a **closed-loop alert system** triggered by event signatures — demonstrated command/response control logic with measurable latency from detection to alert, directly analogous to fault monitoring pipelines in safety-critical embedded systems.

Heterogeneous Integration Review | CIFAR-10 RGB vs. Grayscale CNN | EV Power Module Packaging

Jan – May 2025 · ASU

- **Quantified integration tradeoffs across three industries** by authoring a solo **IEEE-style review** (50+ refs, 2015–2025) on SiC/GaN power electronics, co-packaged optics, and MEMS sensors — benchmarked **NVIDIA CPO at 3.5× power efficiency improvement** over pluggable modules and mapped parasitic-inductance minimization strategies across wirebondless and PCB-embedded die approaches.
- **Isolated color's contribution to CNN classification accuracy** by designing a fully reproducible PyTorch experiment (fixed seeds across Python/NumPy/PyTorch/CUDA, identical model capacity, RGB vs. grayscale only variable) on CIFAR-10 — co-authored EV packaging report quantifying **~75% thermal-resistance reduction** via double-sided cooling and **~½ substrate cost reduction** via ERCD organics vs. alumina DBC.

ACTIVITIES & LEADERSHIP

Community Outreach & Education Volunteer

Aug 2022 – Jan 2024

- Taught professional English and delivered **career-guidance sessions** to underprivileged students; assisted in rural educational outreach programs supporting academic development across underserved communities.
- Conducted motivational sessions on academic pathways and discipline; served as **NSS volunteer** during a one-week residential camp focused on hygiene awareness and student development — demonstrating structured leadership and communication under a defined program mandate.